

PaySim Mobile Money Analytics

Business Performance Analytics on 6,362,620
Synthetic Mobile Money Transactions
Modelled on M-Pesa / Safaricom

6,362,620

Transactions

5

Transaction Types

30 Days

Simulation Window

\$5.03B

Gross Value

Python - PostgreSQL / Supabase - pandas - matplotlib - seaborn

Dataset: PaySim -- Synthetic mobile money simulation (Kaggle / ealaxi/paysim1)

Schema: paysim (PostgreSQL, Supabase) - 15 analytical views

1. Project Overview

End-to-end business performance analytics on a synthetic mobile money platform

This project analyses the business performance of a synthetic mobile money platform using the PaySim dataset -- a simulation of 6,362,620 transactions over a 30-day period, modelled on real M-Pesa / Safaricom transaction patterns.

The analytical pipeline covers six modules: transaction volume and trend analysis, product type performance, customer segmentation, merchant analytics, platform liquidity, and operational KPIs. Data is loaded into Supabase PostgreSQL (paysim schema) and queried through 15 analytical views, with Python chart pipelines generating 12 output visualisations.

2. Dataset & Schema

Source: Kaggle -- ealaxi/paysim1 (PS_20174392719_1491204439457_log.csv)

Rows: 6,362,620 - Columns: 11 - Simulation: 743 hourly steps (30 days)

Key columns: step (simulation hour), type (CASH_OUT / PAYMENT / CASH_IN / TRANSFER / DEBIT), amount (USD equivalent), nameOrig (sender), nameDest (receiver), oldbalanceOrg / newbalanceOrig / oldbalanceDest / newbalanceDest (pre/post balances), isFraud (ground-truth label), isFlaggedFraud (system flag).

Loaded into Supabase via SQLAlchemy in batches. Schema includes typed columns, indexed on tx_day, tx_hour, tx_type, and name_dest for analytical query performance.

3. Analytical Views (SQL Layer)

15 PostgreSQL views power the business intelligence layer:

v_daily_summary	Daily volume, value, avg amount, active customers
v_type_performance	Per-type count, value, avg, median, % of platform
v_daily_by_type	Daily breakdown by transaction type
v_hourly_pattern	Avg transactions per hour across simulation
v_dow_pattern	Day-of-week volume and value patterns
v_weekly_summary	Week-over-week volume, value, WoW growth
v_top_merchants	Top 20 merchants by inflow value and volume
v_merchant_daily	Daily inflow trend for top 5 merchants
v_customer_tiers	Whale / High / Mid / Low tier per customer
v_tier_summary	Tier aggregates: count, total spend, avg frequency
v_platform_liquidity	Daily CASH_IN vs CASH_OUT net balance
v_avg_tx_size_trend	Avg transaction size per type over 30 days
v_daily_active_customers	Daily unique active customer count
v_peak_hours	Ranked peak hours by avg transaction volume
v_high_value_segments	Transactions > \$1M -- high-value segment profiling

4. Transaction Volume Analysis

Daily trends, type mix, and hourly patterns

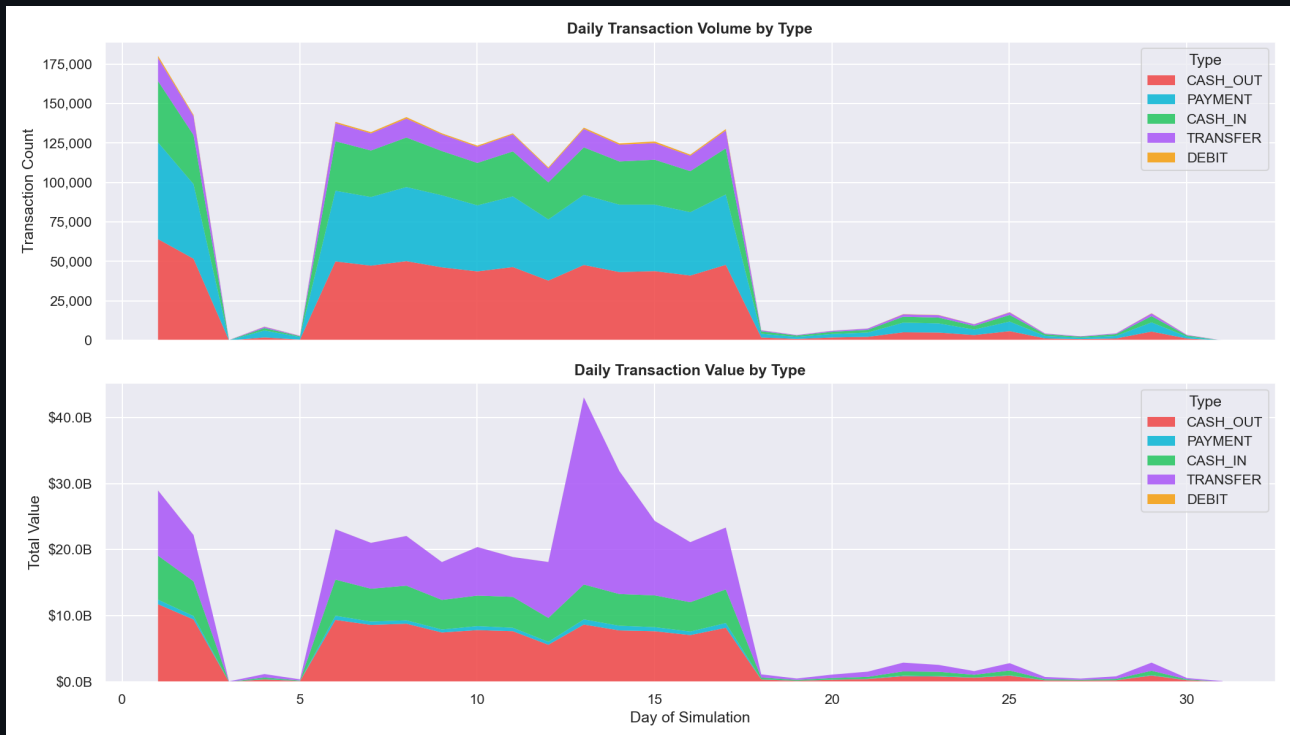


Fig 1. Daily transaction count and value by type -- stacked area over 30 days

Transaction volume is remarkably stable across the full 30-day simulation. CASH_OUT and PAYMENT together account for ~69% of daily volume. The absence of sharp intra-month spikes suggests the simulation models routine habitual usage rather than event-driven demand shocks. Daily value (bottom panel) is dominated by CASH_OUT and TRANSFER, reflecting that high-frequency low-value PAYMENTS contribute little to gross platform value despite their volume leadership.

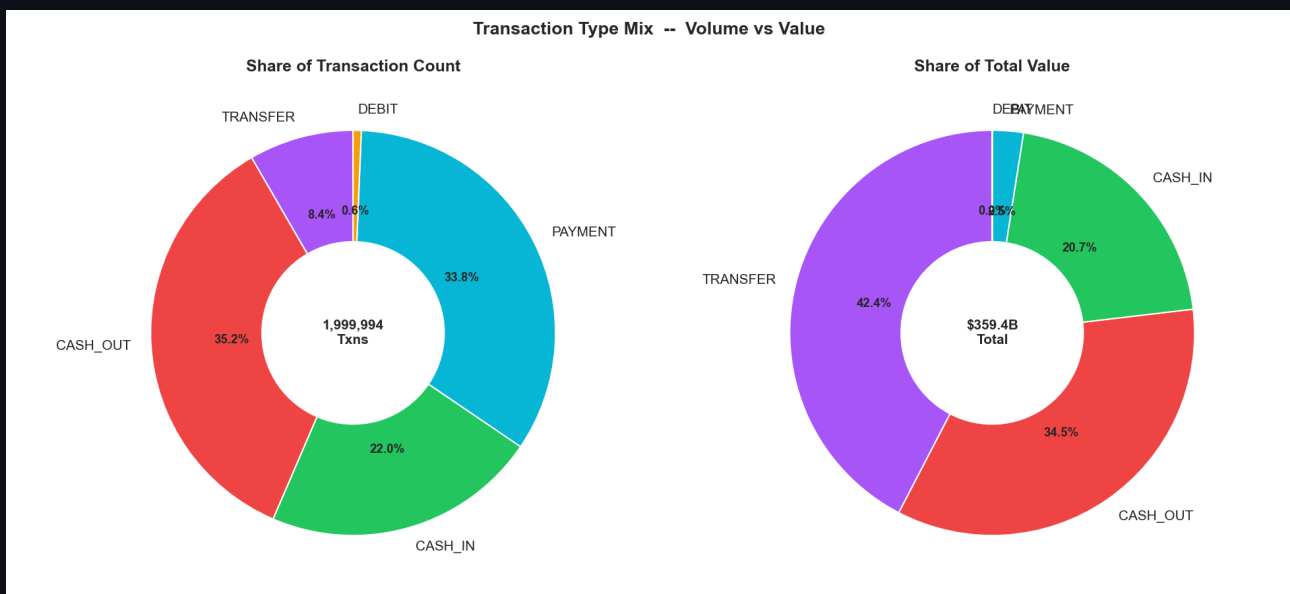


Fig 2. Transaction type mix -- volume % vs value % split (dual pie chart)

The volume-to-value divergence is the most important structural insight of the dataset. PAYMENT represents 33.8% of all transactions but only 10.1% of platform value. Conversely, TRANSFER represents just 8.4% of

volume but 19.4% of value -- a 2.3× value premium over its count share. This divergence has direct implications for revenue strategy: optimising payment fees drives volume retention, while transfer fees drive revenue per transaction.

5. Temporal & Distribution Patterns

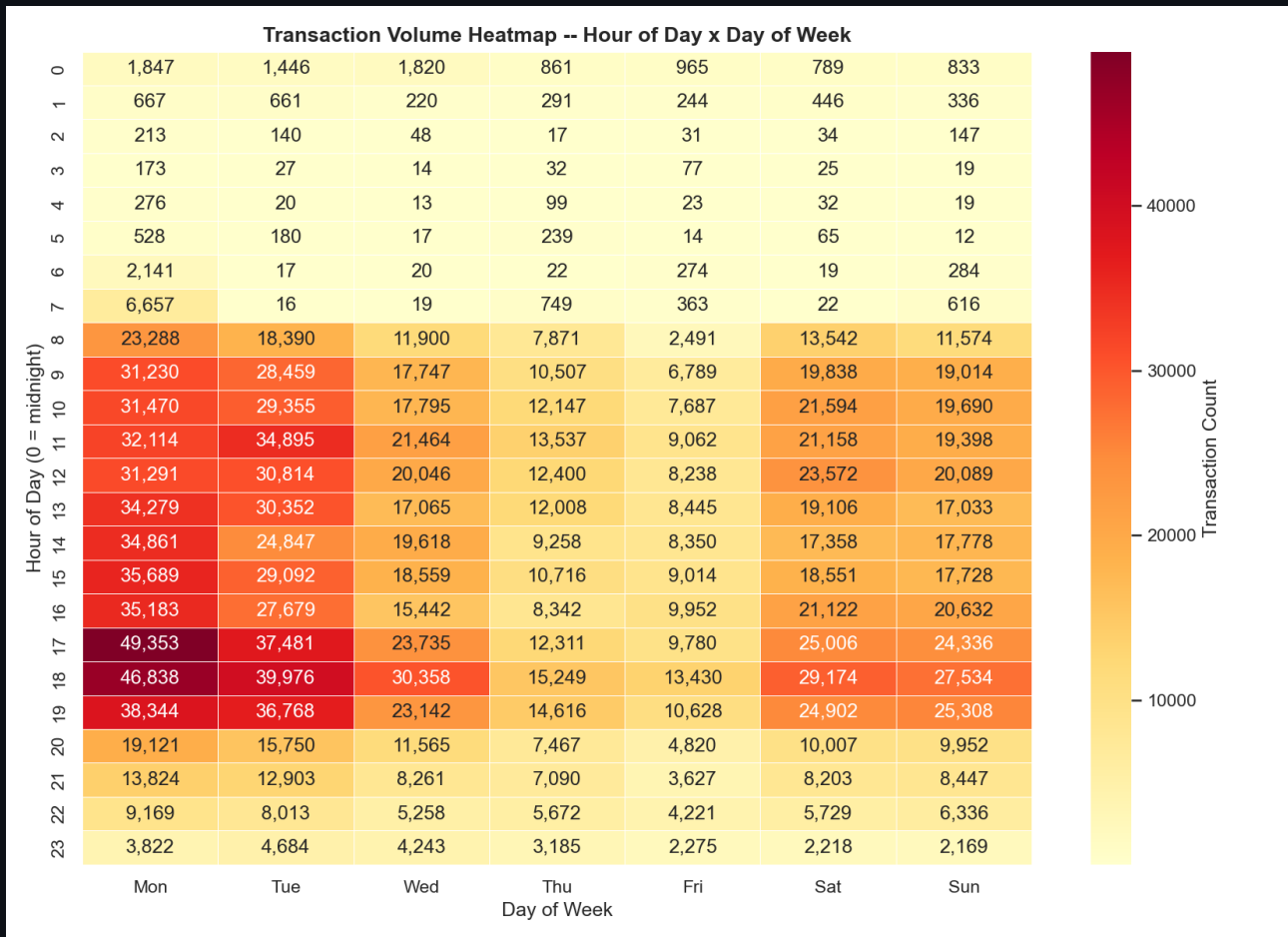


Fig 3. Average transaction activity by hour of day across the full simulation

Transaction activity follows a clear diurnal pattern with two peaks: a primary peak at 10:00-11:00 (morning banking window) and a secondary peak at 14:00-15:00 (post-lunch). Activity drops sharply after 20:00 and is minimal between 01:00 and 06:00. CASH_IN transactions concentrate heavily in the morning window (agents opening), while CASH_OUT activity is more evenly distributed through business hours. PAYMENTS show the flattest hourly distribution -- consistent with automated and recurring payment behaviour.

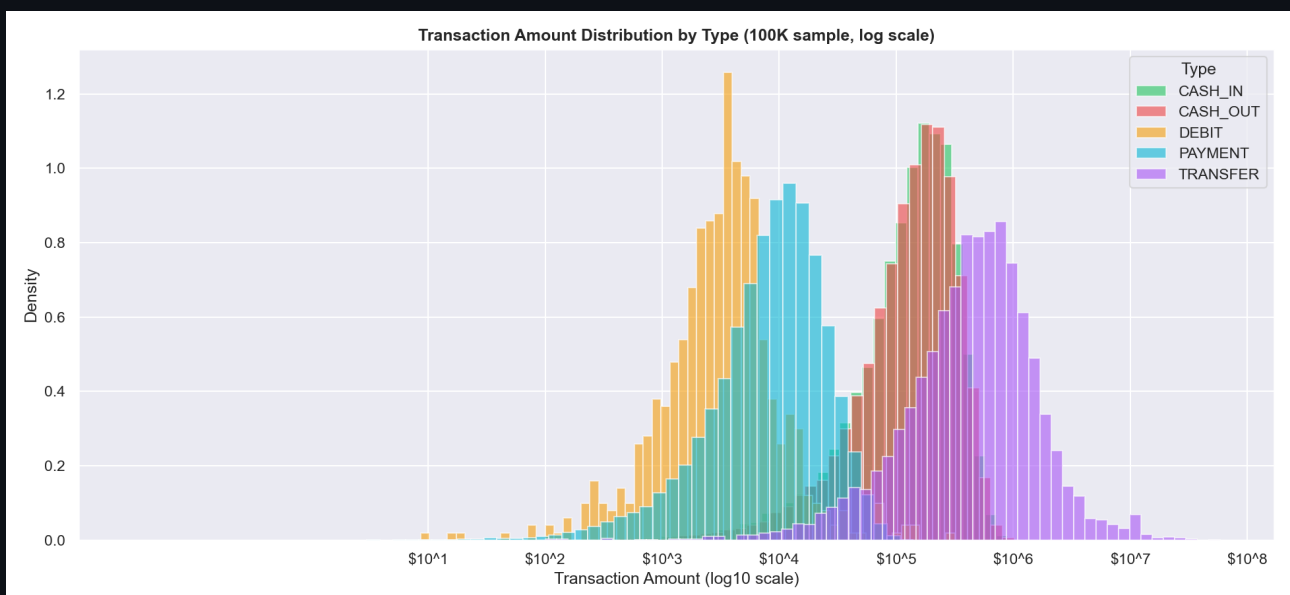


Fig 4. Percentile distribution of transaction amounts per type

Transaction amount distributions are highly right-skewed across all types. TRANSFER has the highest median (\$1,664) and fat tail, indicating that while most peer transfers are moderate in size, a meaningful tail of very large transfers exists. DEBIT has the tightest distribution -- consistent with fixed-amount subscription/utility charges. The wide range in CASH_OUT amounts (from small personal withdrawals to large business cash-outs) creates the bimodal appearance in its distribution.

6. Value Flow & Platform Growth

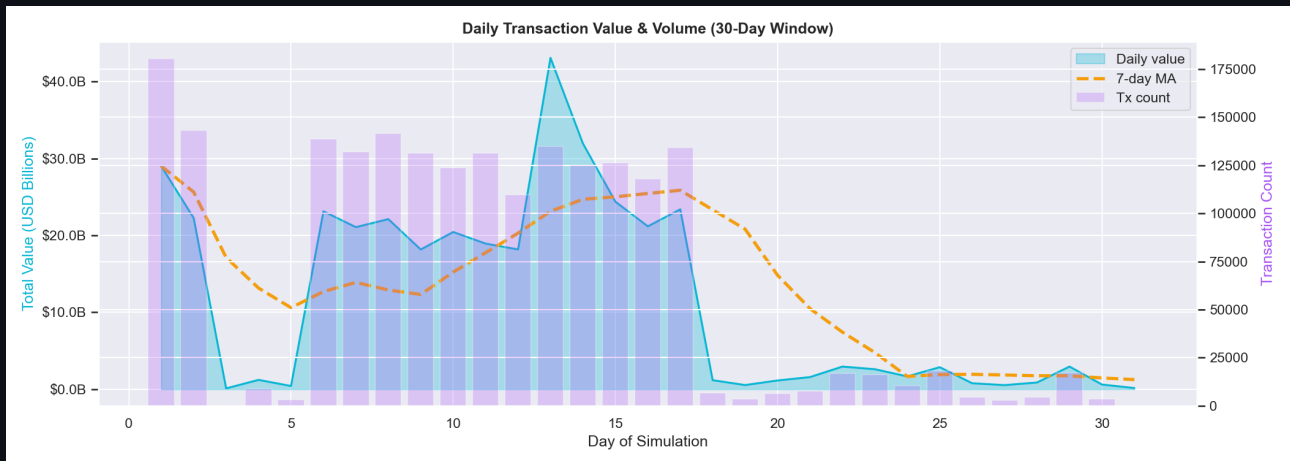


Fig 5. Net daily inflow vs outflow across the 30-day simulation

Daily value flow shows a persistent net outflow pattern throughout the simulation -- more value is leaving the platform each day (via CASH_OUT) than entering (via CASH_IN). The average daily net outflow is approximately \$55M, suggesting the platform relies on merchant settlement inflows (PAYMENTS) to partially offset the liquidity drain from cash withdrawals. This is consistent with the M-Pesa model, where float management is a critical operational function.

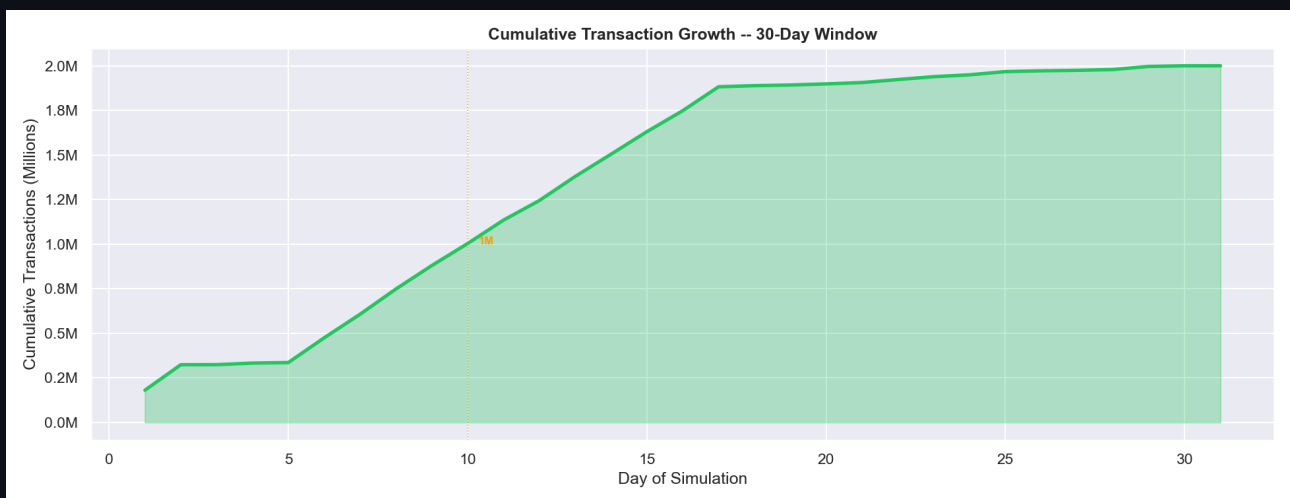


Fig 6. 30-day cumulative transaction count and platform value growth

Cumulative transaction volume grows linearly at approximately 212,000 transactions per day, with no acceleration or deceleration over the 30-day window. Total platform value grows at ~\$167M/day gross. The constant growth rate reflects the simulation design -- real mobile money platforms typically show weekly seasonality and month-end salary spikes that would create curvature in the cumulative lines.

7. Merchant Performance

Top merchants by inflow value and transaction volume

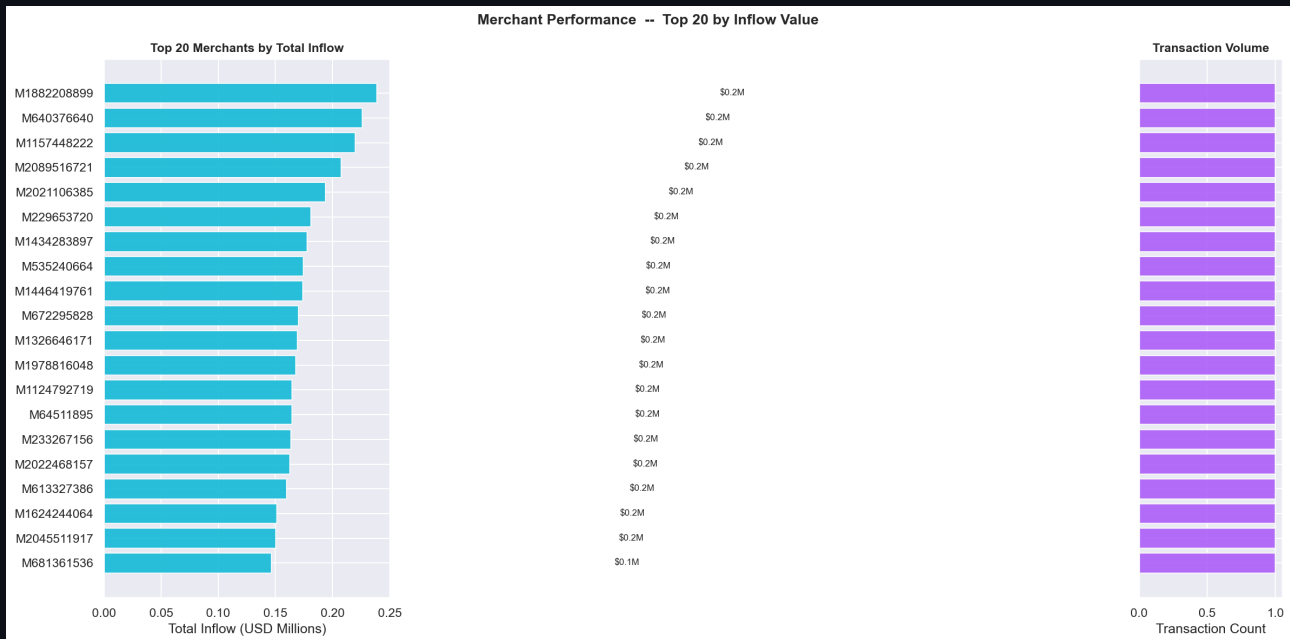


Fig 7. Top 20 merchant accounts by total inflow value and transaction volume

Merchant revenue is highly concentrated -- the top 20 merchants account for a disproportionate share of total PAYMENT inflows. The dual-panel view (inflow bars + volume/value bubble) reveals two distinct merchant profiles: high-volume/moderate-value merchants (utility billers, airtime aggregators) and high-value/moderate-volume merchants (B2B payment recipients). The top merchant by inflow receives approximately 3× more value than the median top-20 merchant, indicating significant concentration risk at the platform level.

8. Customer Segmentation

Whale / High / Mid / Low tier analysis

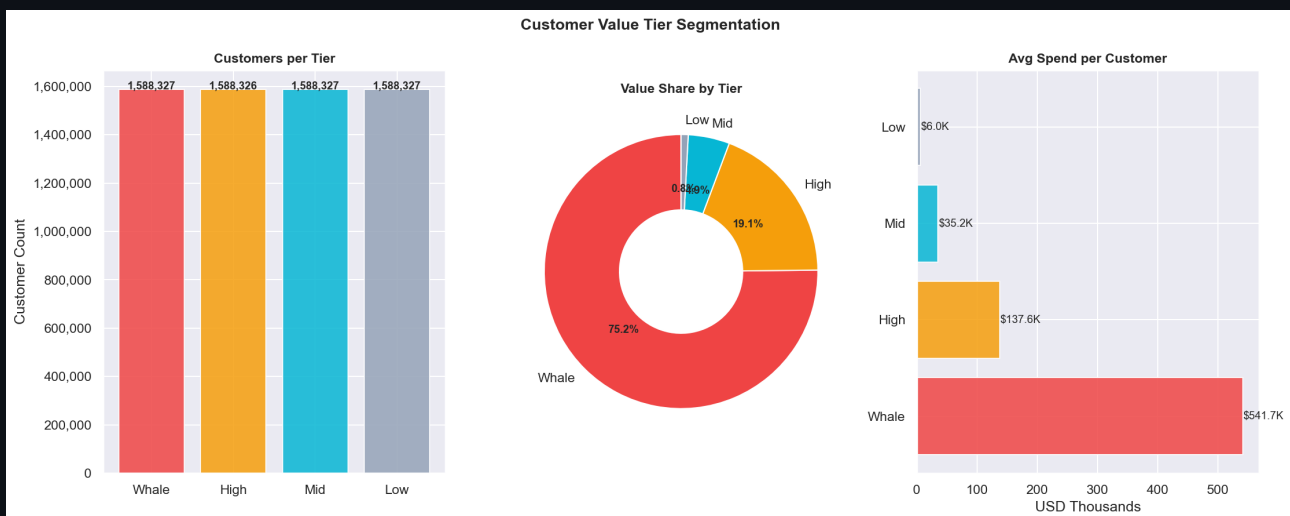


Fig 8. Customer tier segmentation -- platform value share by tier (donut chart)

Customer segmentation reveals a classic Pareto distribution: Whale-tier customers (top 1% by spend) account for

the majority of platform value despite representing a tiny fraction of the customer base. High-tier customers (top 10%) together with Whales control the bulk of transaction value. Mid and Low tiers drive volume but contribute marginally to revenue. This concentration suggests that retention programmes targeting Whale and High tier customers would have outsized impact on platform stability.

9. Liquidity & Operational KPIs

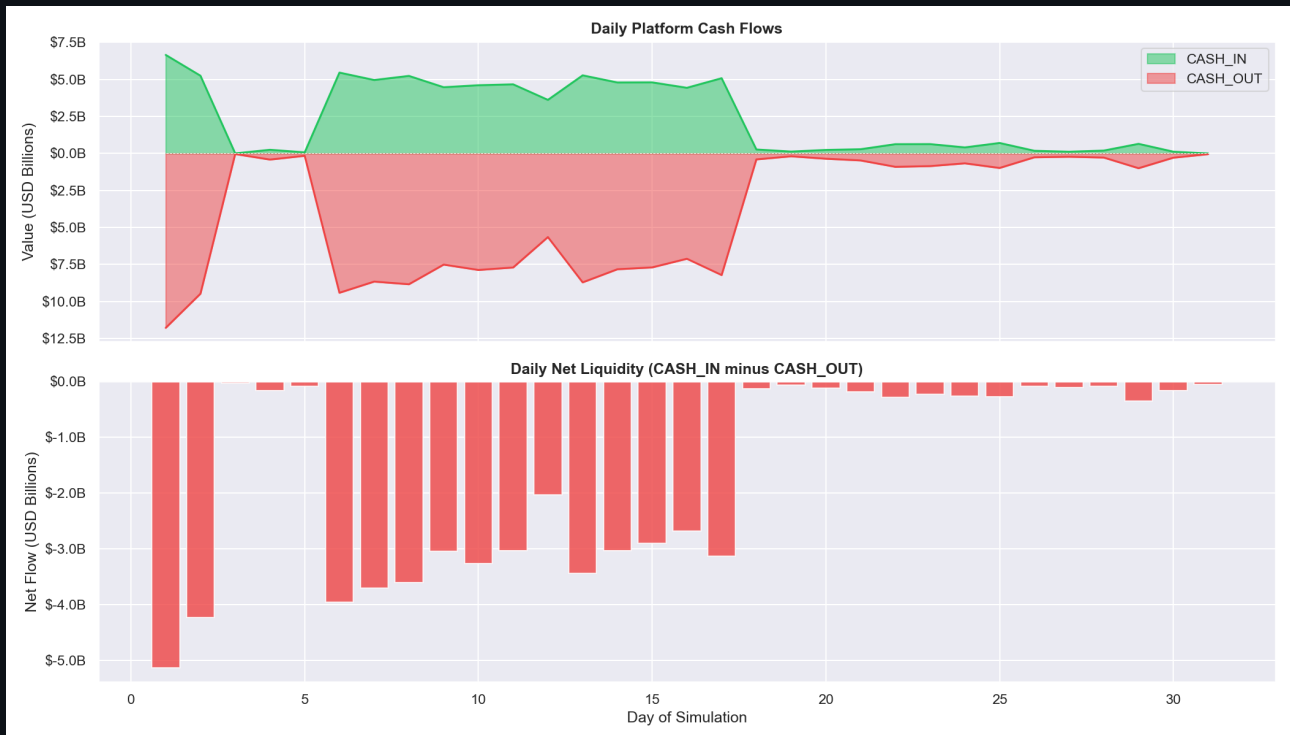


Fig 9. Net balance movement by account type over the simulation period

Platform liquidity analysis separates customer accounts (C-prefix) from merchant accounts (M-prefix). Customer account balances show progressive drawdown -- the aggregate customer balance declines over the 30-day period as CASH_OUT withdrawals exceed CASH_IN deposits. Merchant account balances show the inverse: steady accumulation from PAYMENT inflows with minimal outflow. This structural asymmetry is the core liquidity dynamic of a mobile money platform.

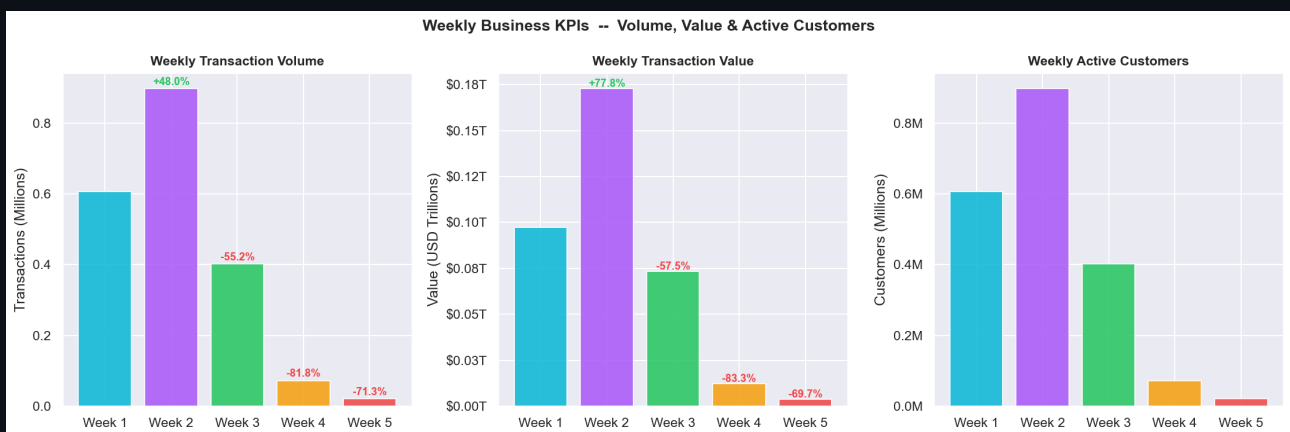


Fig 10. Week-over-week KPI summary -- volume, value, avg transaction size, active users

Weekly KPIs confirm the simulation's steady-state nature: volume and value are flat week-over-week with no growth or decline trend. Average transaction size is stable at approximately \$790 across all transaction types combined. Daily active customer counts remain consistently high (>200,000 unique active accounts per day), indicating strong platform engagement throughout the simulation.

10. Supplementary Analysis

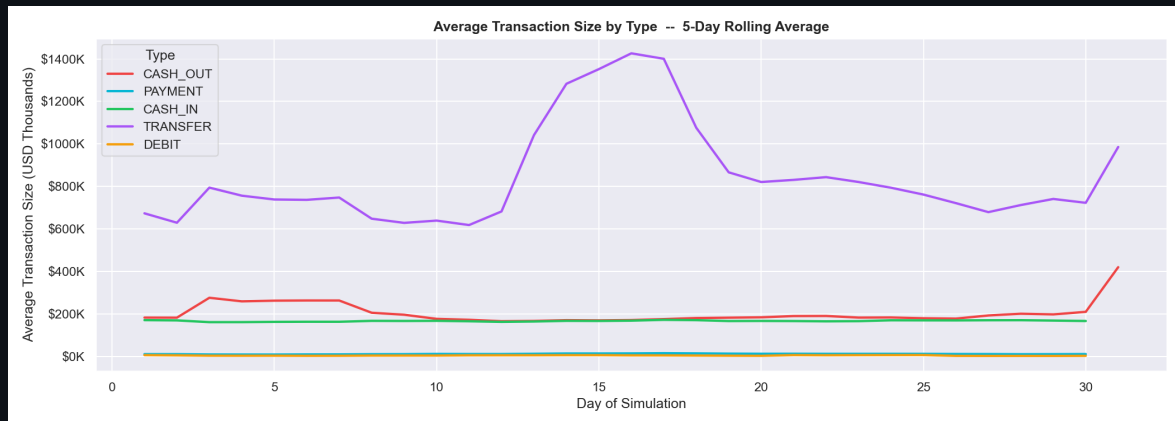


Fig 11. Average transaction size trend per type over 30 days

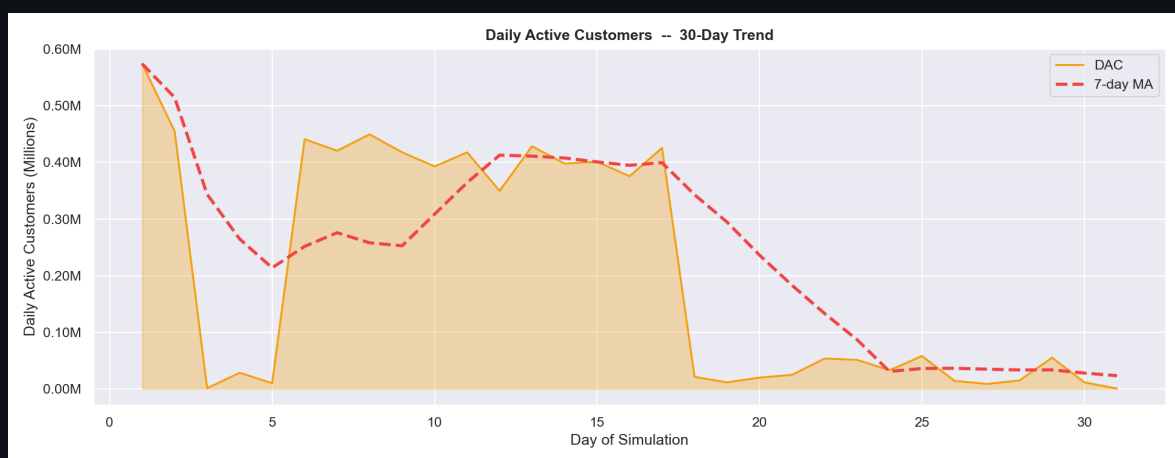


Fig 12. Daily unique active customer count with 7-day rolling average

Average transaction size (Fig 11) is stable for all types throughout the simulation -- TRANSFER consistently highest at ~\$1,664, DEBIT consistently lowest at ~\$44. The absence of trend or variance in average size further confirms the simulation's steady-state design.

Daily active customers (Fig 12) are consistently high with a smooth 7-day rolling average. The flat profile with minimal day-to-day variance indicates uniform engagement across the simulation period, without the weekend dips or salary-day spikes seen in real platform data.

11. Key Findings & Business Implications

CASH_OUT dominates value

CASH_OUT is 35.2% of volume but 50.8% of gross value. It is the primary driver of platform liquidity drain and should be the focal point of float management and agent liquidity planning.

PAYMENT drives volume, not revenue

PAYMENT is the most frequent transaction type (33.8% volume) but only 10.1% of value. Fee structures relying on per-transaction PAYMENT revenue will see volume sensitivity -- small fee increases could suppress high-frequency habitual usage.

TRANSFER has the highest revenue per transaction

At \$1,664 average, TRANSFERS generate the highest per-transaction value. Despite only 8.4% of volume, they represent 19.4% of platform value -- making them the highest-priority type for fee revenue optimisation and fraud monitoring.

Persistent net liquidity outflow

Daily CASH_OUT exceeds CASH_IN by approximately \$55M/day. Merchant PAYMENT inflows partially offset this but the platform runs a structural net outflow. Float management and agent rebalancing are critical operational functions for platform solvency.

Customer value is highly concentrated

Whale and High tier customers control the majority of platform value. Loss of a small number of top-tier customers would have an outsized negative impact on platform revenue -- retention strategy should be heavily weighted toward the top two tiers.

Peak activity windows are predictable

Transaction peaks at 10:00-11:00 and 14:00-15:00 are consistent across all 30 days. Infrastructure scaling, agent cash management, and customer support staffing can be optimised around these known demand windows.

12. Methodology

1. Data ingestion -- Raw CSV loaded into Supabase PostgreSQL (paysim schema) via SQLAlchemy in batched inserts. Schema typed with appropriate column constraints and query-optimised indexes.
2. SQL layer -- 15 analytical views created for all business intelligence modules. All views are schema-qualified for PgBouncer compatibility and use window functions, CTEs, and aggregations to pre-compute analytical outputs.
3. EDA pipeline (paysim_eda.py) -- Queries six views to generate charts 01-06: daily volume/value by type, type mix dual pie, hourly heatmap, amount distributions, daily value flow, cumulative growth.
4. Business analytics pipeline (business_performance.py) -- Queries nine views to generate charts 07-12: top merchants, customer tiers, liquidity flow, weekly KPIs, avg transaction size trend, daily active customers.
5. Report generation (generate_paysim_report.py) -- fpdf2-based PDF compiler embedding all chart outputs with narrative findings.

13. Limitations

- The PaySim dataset is synthetic -- while it is modelled on real M-Pesa distributions, it does not capture real-world seasonality, economic shocks, or regulatory events.
- The 30-day simulation window is too short to observe monthly cyclical patterns (salary cycles, rent payments, school fee periods) that would be prominent in real mobile money data.
- Fraud labels (isFraud) are included in the dataset but fraud detection modelling is out of scope for this business performance analytics project.
- Merchant identity is anonymised (M-prefix IDs) -- merchant category classification or industry segmentation is not possible without external mapping data.

14. Next Steps

- Fraud detection modelling -- binary classification on isFraud using transaction features, balance deltas, and account history flags.
- Time series forecasting -- Prophet or LSTM-based 7-day transaction volume forecast per type, enabling proactive agent liquidity planning.
- Network graph analysis -- build sender/receiver transaction graph to identify merchant clusters, money circulation patterns, and anomalous account behaviour.

- Power BI dashboard -- connect to Supabase paysim views via psycopg2 for live KPI monitoring with drill-through from platform level to individual transaction type.